CS425: Computer Networks
Homework 1

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Question 1: CRC Frame Generation and Error Detection

A program was implemented to generate an n-bit frame from a k-bit data block D and a $(n-k+1)$-bit CRC pattern P using modulo-2 division.

Working Procedure:

- Accept input data bits D and generator polynomial P.
- Append $(|P|-1)$ zeros to D.
- Perform binary division using XOR operations.
- Append the remainder to form the transmitted frame T.

The program was further modified to:

- Generate a random 10-bit message.
- Use $P = 110101$ to generate a 15-bit transmission frame.
- Introduce transmission errors at arbitrary bit positions.
- Apply CRC at the receiver to determine acceptance or rejection.

The experimental outputs verified correct CRC generation and successful error detection.

Question 2: Go-Back-N Window Size

In Go-Back-N ARQ, sequence numbers are represented using k bits, allowing 2^k distinct sequence numbers.

If the sender window size were 2^k, ambiguity would occur after sequence number wraparound. The receiver would be unable to distinguish between retransmitted old frames and newly transmitted frames with identical sequence numbers.

To avoid this ambiguity, the sender window size is limited to:

$$W = 2^k - 1$$

This guarantees that at least one sequence number remains unused, ensuring reliable communication.

Question 3: Selective-Reject Window Size

In Selective-Reject ARQ, the receiver accepts and buffers out-of-order frames.

Because sequence numbers wrap around after 2^k, delayed frames may overlap with new frames if the window size is too large.

To prevent overlap between old and new frames, the maximum allowable window size is:

$$W_{\max} = 2^{k-1}$$

This ensures no ambiguity occurs between retransmitted and newly transmitted frames.

Question 4: Stop-and-Wait Efficiency

The efficiency of the Stop-and-Wait protocol is:

$$\eta = \frac{T_t}{T_t + 2T_p}$$

Given:

$$R = 4 \text{ kbps} = 4000 \text{ bits/sec}$$

$$T_p = 20 \text{ ms} = 0.02 \text{ sec}$$

Transmission time:

$$T_t = \frac{L}{R} = \frac{L}{4000}$$

Thus,

$$\eta = \frac{L/4000}{L/4000 + 0.04}$$

For efficiency $\eta \geq 0.5$:

$$\frac{L}{4000} \geq 0.5 \left(\frac{L}{4000} + 0.04 \right)$$

Solving:

$$L \geq 160 \text{ bits}$$

Therefore, Stop-and-Wait achieves at least 50% efficiency when:

$$\boxed{L \geq 160 \text{ bits}}$$

Question 5: Frame Error Probability

Given bit error probability:

$$P_b = 10^{-3}$$

Each bit is correct with probability:

$$1 - P_b = 0.999$$

(a) Probability of No Errors

$$P(\text{no error}) = (0.999)^4 = 0.996006$$

(b) Probability of At Least One Error

$$P(\text{at least one error}) = 1 - 0.996006 = 0.003994$$

(c) Probability of Undetected Error with Parity

With one parity bit, total bits = 5.
Parity fails when an even number of bits are corrupted.
Probability of exactly two errors:

$$P(2) = \binom{5}{2}(10^{-3})^2(0.999)^3 \approx 9.97 \times 10^{-6}$$

Higher-order even errors are negligible.

$$\boxed{P(\text{undetected}) \approx 1 \times 10^{-5}}$$

Question 6: CRC Calculation

Given:

$$P = 110011, \quad M = 11100011$$

Since generator length is 6, append 5 zeros:

$$11100011\,00000$$

Modulo-2 division yields remainder:

$$R = 11010$$

Final transmitted frame:

$$11100011\,11010$$

Verification division confirms zero remainder.

Question 7: CRC Encoding and Error Detection

Generator polynomial:

$$P(x) = x^4 + x + 1$$

Binary representation:

$$P = 10011$$

(a) Encoding

Message:

$$M = 10010011011$$

Appending four zeros:

$$10010011011\,0000$$

Remainder obtained:

$$R = 1100$$

Encoded frame:

$$10010011011\,1100$$

(b) Error Pattern: 100010000000000

The received frame is obtained by XORing the encoded frame with the error pattern. When this received frame is divided by the generator, the remainder is nonzero. Since the remainder is not zero, the receiver detects the error. Therefore, the corruption is successfully detected.

(c) Error Pattern: 100110000000000

Applying this error pattern and dividing the resulting frame by the generator produces a zero remainder. Because the remainder is zero, the error is not detected. This shows that CRC cannot detect all possible error patterns, although such undetected cases are rare.